

The Klein-Gordon Equation

Kieron Eisma

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Introduction

This article provides a brief derivation of the Klein-Gordon (or Klein-Gordon-Fock) Equation for a free particle by substituting the corresponding energy and momentum operators into the relativistic energy-momentum relation.

Email:

`kiwi.eisma@gmail.com`

Derivation

We recall that the energy of a relativistic free particle is given by

$$E^2 = m_0^2 c^4 + p^2 c^2$$

where m_0 denotes the particle's rest mass. Recall that the corresponding energy and momentum operators are given by

$$\hat{E} = i\hbar \frac{\partial}{\partial t}$$

and

$$\hat{p} = -i\hbar \nabla$$

So substituting into the energy-momentum relation above yields

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = m_0^2 c^4 \psi - \hbar^2 c^2 \nabla^2 \psi.$$

Rearranging yields

$$\begin{aligned}\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} - \nabla^2 \psi + \frac{m_0^2 c^2}{\hbar^2} \psi &= 0 \\ \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m_0^2 c^2}{\hbar^2} \right) \psi &= 0.\end{aligned}$$

If we define the d'Alembertian operator

$$\square = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2,$$

the equation above simplifies to yield

$$\left(\square + \frac{m_0^2 c^2}{\hbar^2} \right) \psi = 0,$$

which is a clean form of the Klein-Gordon Equation.

One-Dimensional Solution

Consider a free particle moving along the x -axis. The Klein-Gordon Equation simplifies to

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} + \frac{m_0^2 c^2}{\hbar^2} \right) \psi = 0.$$

We consider plane wave solutions $\psi(x, t) = Ae^{i(kx - \omega t)}$. First, note that taking partial derivatives shows¹

$$\partial_t^2 \psi = -\omega^2 \psi \qquad \partial_x^2 \psi = -k^2 \psi.$$

Hence, we require

$$-\frac{\omega^2}{c^2} \psi = -k^2 \psi - \frac{m_0^2}{\hbar^2} c^2 \psi,$$

that is, $\frac{\omega^2}{c^2} = k^2 + \frac{m_0^2}{\hbar^2}$.

Rearranging and taking the square root gives

¹I adopt the notation $\partial_t = \frac{\partial}{\partial t}$, and similarly for ∂_x .

$$\omega = \pm \sqrt{c^2 k^2 + \frac{c^4 m_0^2}{\hbar^2}}.$$

From the de Broglie relation, $p = \hbar k$.

$$\omega = \pm \frac{1}{\hbar} \sqrt{(pc)^2 + (m_0 c^2)^2}.$$

But the expression on the right is just the relativistic energy:

$$\omega = \pm \frac{1}{\hbar} E.$$

This yields two solutions, one for positive frequency ($\omega = \frac{1}{\hbar} E$) and the other corresponding to negative frequency ($\omega = -\frac{1}{\hbar} E$):

$$\begin{aligned} \psi_1(x, t) &= A e^{i(kx - \frac{1}{\hbar} Et)} \\ &= A e^{\frac{i}{\hbar} (px - Et)}, \end{aligned}$$

and

$$\psi_2(x, t) = B e^{\frac{i}{\hbar} (px + Et)}$$

because $p = \hbar k$.